

COSC 341: Human Computer Interaction

Lab 3: Quiz Application (20 marks)

The purpose of this lab is to learn how to create Android multi-view applications, Android Intents, search view, and list view.

Configuration note:

- Please use Android Studio (with Java as the programming language—recommended platform for this course). The TAs will use this to load, build, and run your code. If you use any other toolkit or software development kit (e.g., Flutter), please write the corresponding name of the toolkit in the submission comment box in Canvas.
- Assignments will be marked using an Android Virtual Device (AVD) running a **Pixel 3a (API 33 or 35)** is preferred).

Submission note:

Create a folder with your student number. Then, create two subfolders, **Code** and **APK**, under the folder.

- The **Code** subfolder should include your Android project code for questions 1 and 2
- **APK** folder should contain two .apk files – one for question 1 and another for question 2.

Now, create a Zip file of the folder with your student number and upload it to Canvas.

Also, in the Canvas submission comment box, please include **two video links (one for question 1 and another for question 2) showing** the app working for different user inputs.

Question 1 [15 marks]: In this question, you will implement an Android quiz application. You have freedom in designing the interface screens (e.g., buttons vs. image buttons and typography, colour, etc), and select strategies (e.g., multiple activities, fragments).

Requirements:

[1.5 marks] Welcome Screen

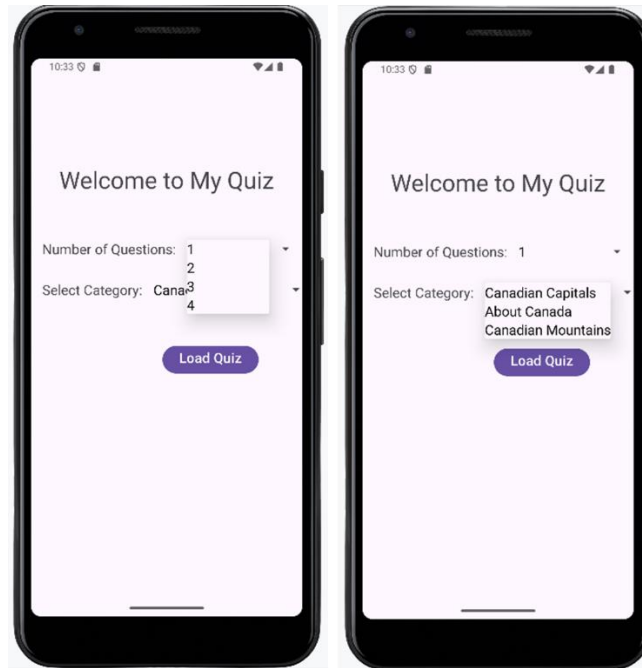
Your application should include a Welcome screen. This screen includes:

[0.5 mark] A TextView showing the number of questions and a spinner showing values between 1 and 4. A user can select any number between 1 and 4.

[0.5 mark] A TextView showing the question categories and a spinner showing the question categories. There should be at least 3 question categories. For example, one question category can be on the “Canadian Capital”, one can be on “About Canada” and another can be on “Canadian mountains”. Feel free to come up with your own categories and four questions under each category.

[0.5 mark] A button to navigate to the next screen (e.g., the “Load Quiz” button in the image). When the user clicks on this button, the question screen opens.

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[3 marks] Questions Screen(s) × Three Categories = 9 marks

A question screen shows a question, an image, and options. The questions should be loaded based on the category selected in the previous screen.

[1 mark] There should be n number of screens where n is the number of questions selected in the *Welcome Screen*. A user can't proceed to the next question without selecting an option.

[1 mark] A TextView to show a question, an ImageView to show the image related to the question, four options with four radio buttons, and a button (e.g., "Next") to move to the next question.

[1 mark] When the user reaches the last question, the button text should be "Finish". Clicking on the "Finish" button opens the Results Screen.

[1 mark] Results Screen × Three Categories = 3 marks

This interface shows the scores based on user responses to the questions. It includes:

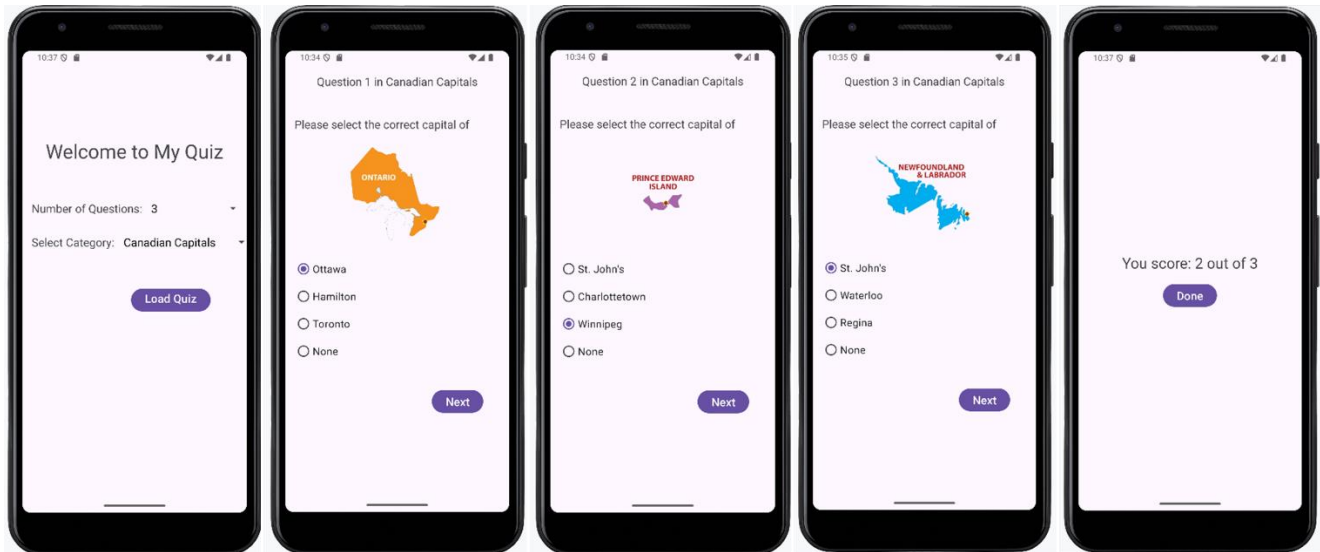
[0.5 mark] a TextView showing the correct score. Each question is worth one mark.

[0.5 mark] A button (e.g., "Done") to go back to the Welcome screen. Once in the Welcome screen, the user begins the quiz. If the user retakes the quiz, the application shouldn't show any previously selected responses (e.g., response to question 1 in the previous attempt).

[1.5 mark] A video showing how the app works.

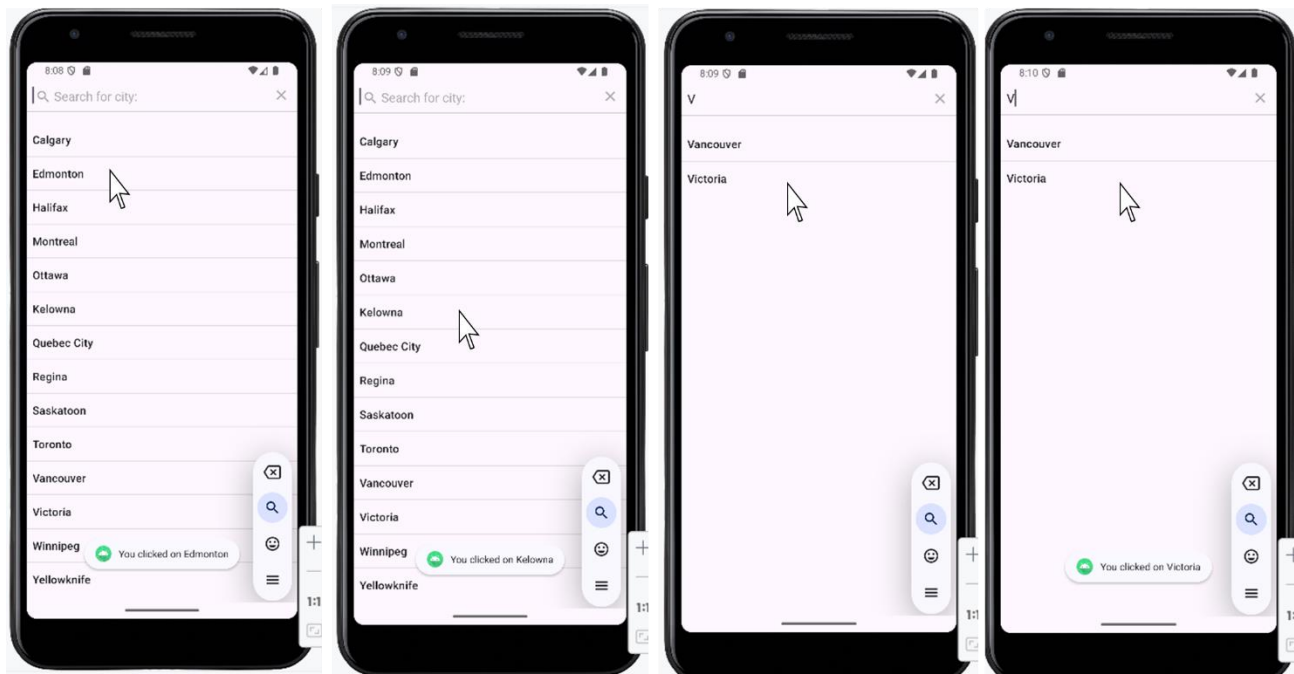
The video should show the results screen for all 3 categories for 4 questions with (i) all correct answers and (ii) at least one wrong answer. The screenshot below is for "1 category for 3 questions with one wrong answer."

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Question 2 [5 marks]:

Create an Android Application with a ListView and SearchView:



[2 mark] ListView And SearchView:

- Add a SearchView on top of an Empty View Activity.
- Below the SearchView, add a ListView for users to search through the list of cities. Please add SearchView query hint saying, "Search for city: "
- Add 15 Canadian cities and display them in the ListView.

[2 marks] Implement the Search Feature:

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- When a user clicks on any city from the ListView, show a Toast message displaying the name of the clicked city.
- Allow users to filter the city list by typing in the SearchView. As the user types, the list of cities should update dynamically to match the search query. Also, display a Toast When a City is Clicked from the filtered list

[1 mark] A video showing how the app works.

Other Grading Criteria

[-2 mark] Source code is very sparsely commented, poorly formatted (indent, spacing)

[up to -20] Submission instruction not followed (e.g., no APKs)

Resources:

1. Use Android SearchView
<https://developer.android.com/reference/android/widget/SearchView>
2. Use Android ListView:
<https://developer.android.com/reference/android/widget/ListView>